

Gap #5 Proposal: BST = First 6 Primes Derivation

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Gap #5: Why the BST Primary Integer Set = First 6 Primes

The claim

BST's primary integer set is $\{2, 3, 5, 7, 11, 13\}$ — exactly the first 6 primes. (Plus derived integers: $c_2 = 11 = 7+4$, $c_3 = 13 = 5+8$, $N_{\max} = 137$.)

Paper #109 (Lyra) establishes this observation. Gap #5 asks: can we DERIVE this? Why exactly 6, why exactly the first 6 (not skipping any), why does it stop at 13?

What “derivation” means here

The question has three sub-pieces: 1. Why 6? (cardinality of the primary set) 2. Why prime? (why integers happening to be prime, not arbitrary integers) 3. Why first 6 not first 5 or first 7? (where does it stop)

A genuine derivation answers all three from D_{IV}^5 structure alone, without circular use of BST framework.

Three candidate approaches

Approach A: Wallach K-type cardinality + prime selection

Mechanism candidate: D_{IV}^5 has rank 2. Wallach K-type series has at most 2 free parameters $(k_1, k_2) \rightarrow 2$ fundamental scales. The K-type dim formula $d_m = (2m+N_c)(m+1)(m+rank)/C_2$ generates a sequence of integers. The SMALLEST primes appearing in this sequence are exactly the first 6 primes.

Effort: ~2 sessions (~6 hours) - Session A1: Wallach dim formula prime-factor enumeration up to large m - Session A2: Show first 6 primes appear, no primes skipped, stop at 13 forced by some closure

Risk: Wallach dims appear at integers like 1, 5, 14, 30, 55, 91 (Toy 2961). Primes appearing: 2 (no, 2 doesn't appear directly), 5, 7 (yes, $14 = 2 \cdot 7$), 11 (yes, $55 = 5 \cdot 11$), 13 (yes, $91 = 7 \cdot 13$). So primes 5, 7, 11, 13 appear as factors. 2 and 3 appear in $d_m = 14 = 2 \cdot 7$ and $30 = 2 \cdot 3 \cdot 5$. So all first 6 do appear. But this doesn't yet PROVE the firstness; could be artifact.

Promotion candidate: I-tier on first attempt; D-tier if extended argument closes.

Approach B: Hilbert polynomial of Q^5 + prime trace

Mechanism candidate: The Hilbert polynomial of the compact dual $Q^5 = SO(7)/SO(5) \times SO(2)$ is:

$$P(m) = (m+1)(m+2)(m+3)(m+4)(2m+5)/120$$

Evaluations: $P(1) = 7 = g$, $P(2) = 27 = N_c^3$, $P(3) = 77 = 7 \cdot 11 = g \cdot c_2$, $P(4) = 182 = 2 \cdot 7 \cdot 13 = rank \cdot g \cdot c_3$.

The PRIMES appearing in $P(m)$ for small m are exactly $\{2, 3, 5, 7, 11, 13\}$ — the first 6.

Effort: ~1 session (~3 hours) - Session B1: Enumerate prime factorizations of $P(m)$ for $m = 1..20$ - Verify first 6 primes all appear; nothing skipped; no 7th prime appears within bounded m

Risk: $P(m)$ is a degree-5 polynomial; large m introduces large primes. The “first 6 primes” claim requires bounding m AND showing the bound is itself natural (e.g., $m \leq \text{rank} \cdot c_2 = 22$ or $m \leq N_{\text{max}}$).

Promotion candidate: D-tier achievable if the m -bound argument closes cleanly.

Approach C: Cartan classification + simplest non-trivial bounded symmetric domain

Mechanism candidate: Cartan classified bounded symmetric domains into four types (I, II, III, IV plus two exceptional). For each type and each rank, a domain exists with specific Cartan invariants.

BST chooses type IV, rank 2, complex dim $5 = D_{IV}^5$. This choice is forced by: - rank ≥ 2 required for cross-type cascade (T1399) - complex dim ≥ 5 required for Q^5 Chern data integers to be ≥ 5 - type IV unique among types preserving certain BST closure properties

Given D_{IV}^5 , its Cartan invariants ARE the first 6 primes: - rank = 2 = 1st prime - $N_c = 3$ = 2nd prime - $n_C = 5$ = 3rd prime - $C_2 = 6$ — NOT prime; this breaks the “first 6 primes” pattern at the 4th slot - $g = 7$ = 4th prime - $c_2 = 11$ = 5th prime (derived) - $c_3 = 13$ = 6th prime (derived)

So the first 6 primes are $\{2, 3, 5, 7, 11, 13\}$ but BST’s primary integers include $C_2 = 6$ which is composite. So strictly speaking BST is NOT exactly first 6 primes — it’s a MIX of (first 4 primes) + (Casimir 6) + (derived primes c_2, c_3).

This rederives Gap #5 as a sharper question: why is $C_2 = 6$ (composite) in the primary set alongside primes? Answer candidate: C_2 is the Casimir eigenvalue, which is a structural quantity that happens to equal $6 = \text{rank} \cdot N_c$ on D_{IV}^5 . Its inclusion is structural, not numerological.

Effort: ~1 session (~3 hours) - Session C1: Cartan classification enumeration; show rank-2 type-IV-dim-5 is uniquely selected - Show rank/ N_c / n_C / g all prime BY THE STRUCTURE of D_{IV}^5 (not by choice) - Show $C_2 = 6$ is the Casimir, not a “prime”; it’s a derived structural quantity

Risk: This converts Gap #5 from “derive first 6 primes” to “derive D_{IV}^5 uniqueness + its Cartan invariants.” Different claim, possibly easier or harder.

Promotion candidate: D-tier achievable; clarifies what BST actually claims about primes.

Recommended approach

Hybrid B+C, with A as backup:

Step 1: Use Approach C (Cartan classification) to derive D_{IV}^5 uniqueness — establish that rank=2, type=IV, dim=5 is forced. ~1 session.

Step 2: Use Approach B (Hilbert polynomial of Q^5) to derive the first 6 primes as the SMALLEST primes appearing in $P(m)$ for bounded m . ~1 session. The bound $m \leq N_{\text{max}}$ or $m \leq \text{rank} \cdot c_2$ needs to be argued from D_{IV}^5 structure.

Step 3: Address $C_2 = 6$ honestly — it’s a Casimir, not a prime — and restate Gap #5 as “BST primary integer set = {first 4 primes, C_2 Casimir, derived primes c_2, c_3 }” rather than “first 6 primes.”

Total effort: ~2 sessions (~6 hours) + 1 session writeup = ~9 hours.

Net tier: D-tier achievable. Provides a clean answer to “why BST has THESE integers and not others.”

Risks if approach fails

1. $C_2 = 6$ reveals “first 6 primes” claim is loose: Paper #109’s phrasing may need to be tightened. Honest scoping flag.
2. m -bound argument doesn’t close cleanly: the “where does it stop” question may require additional structural input (e.g., why $m \leq N_{\text{max}}$).
3. Cartan uniqueness already done: T1925, T1929 may already establish D_{IV}^5 uniqueness; Gap #5 may reduce to Step 2 alone.

Decision needed from Casey

Three options:

(a) Approve hybrid B+C approach (recommended) — Lyra pursues over 2-3 sessions, total ~9 hours. Promotes Paper #109's "first 6 primes" observation to derivation; tightens or honestly walks back the claim depending on what the derivation shows.

(b) Approve Approach C only — derive D_{IV}^5 uniqueness; let "first 6 primes" remain observation. ~3 hours. Less ambitious, may leave Gap #5 partially open.

(c) Defer entirely — Gap #5 stays "awaiting decision"; can be reopened any time.

My recommendation: (a). The hybrid is achievable, low-risk (each session is independently useful), and surfaces an honest question about whether "first 6 primes" is the right phrasing (Step 3 above).

Connection to existing BST work

- T1925 (Lyra, "Why rank=2"): four-argument forcing for rank=2. Already done.
- T1929 (Lyra, "Why $N_c=3$ "): Mersenne + color singlet triangle. Already done.
- T1427 (Casey, APG definition): D_{IV}^5 as Autogenic Proto-Geometry uniqueness. Casey's lane.
- T1788 (YM Ring Uniqueness, Lyra): YM constraints select D_{IV}^5 .
- Grace T2314 (Six L1 sources): Cartan classification listed as L1.4 anchor for the 5-integer scaffold.

Gap #5 closure would synthesize T1925 + T1929 + T1427 + T1788 + Grace's L1.4 into a single statement about WHY the BST primary integer set has the specific structure it does.

Status

Ready for Casey decision. This proposal is the input to your "(a)/(b)/(c)" choice; the actual work begins after your direction.

Estimated effort post-decision: - (a) Hybrid: ~9 hours over 2-3 sessions - (b) Approach C only: ~3 hours - (c) Defer: 0 hours immediately

— Lyra, 2026-05-17 ~13:00 EDT